

Tumors and Tumorlike Conditions of the Anal Canal and Perianal Region: MR Imaging Findings¹

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Abbreviations: FDG = fluorine 18 fluorodeoxyglucose, HIV = human immunodeficiency virus, HPV = human papillomavirus, SCC = squamous cell carcinoma

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the anal canal anatomy and MR imaging techniques for the evaluation of the anal canal and perianal region.
- Explain the epidemiology, classification, and pathogenesis of malignancies involving the anal canal and perianal region.
- Describe the cross-sectional imaging findings and the role of imaging in the diagnosis and surveillance of anal canal and perianal malignancies.

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Tumors and tumorlike conditions of the anus and perianal region originate from the anal canal and anal margin or result from direct extension of tumors from adjacent organs. The anatomy of the anal canal is complex, and its different histologic characteristics can lead to diverse pathologic conditions. The anal canal extends from the anorectal junction to the anal verge. The World Health Organization classification of anal canal tumors includes (a) anal intraepithelial neoplasia, the precursor of squamous cell carcinoma (SCC), and (b) invasive tumors. Invasive tumors are further classified on the basis of cell type as epithelial tumors (SCC, adenocarcinoma, mucinous adenocarcinoma, small cell carcinoma, and undifferentiated carcinoma), nonepithelial tumors, carcinoid tumors, melanoma, and secondary tumors (direct spread from rectal, cervical, or prostate carcinoma). The anal margin, or perianal skin, lies outside the anal verge and encompasses a radius of 5 cm from the anal verge. Tumors in the anal margin are classified according to the World Health Organization classification of skin tumors. Anal margin tumors include SCC, anal intraepithelial neoplasia, also known as Bowen disease, adenocarcinoma and its precursor Paget disease, basal cell carcinoma, and verrucous carcinoma (Buschke-Löwenstein tumor), which is a rare variant of SCC. Imaging plays an important role in the evaluation, staging, and follow-up of patients with anal and perianal tumors. However, because of the overlap in imaging features among these diverse entities, a definitive diagnosis is best established at histopathologic examination. Nevertheless, familiarity with the pathogenesis, imaging features, and treatment of these tumors can aid radiologic diagnosis and guide appropriate patient treatment.

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Introduction

Anal cancer accounts for 2% of all gastrointestinal malignancies and only 0.4% of all new cancers diagnosed annually in the United States (1). Squamous cell carcinoma (SCC) is the most common histologic type of anal canal carcinoma. In the United States, the incidence of SCC has increased, with 3500 new cases in 2001 and 3990 new cases in 2005; for 2015, it was estimated that there would be 7270 new cases of anal cancer and that an estimated 1010 people would die of this disease (2). The number of new cases of anal cancer is 1.8 per 100 000 men and women per year according to 2008–2012 data (2). Anal infection with human papillomavirus (HPV), in particular HPV types 16 and 18, is the most important risk factor for the development of anal SCC (3). HPV causes anal intraepithelial neoplasia, which may progress to invasive cancer (4). In most cases, anal infection with

TEACHING POINTS

- The muscles surrounding the anal canal control continence and can be separated into the internal anal sphincter and the external anal sphincter. The internal anal sphincter is the direct extension of the inner circular smooth muscle layer of the rectum and is innervated by both sympathetic and parasympathetic nerves, allowing for a state of tonic contraction and baseline continence. At magnetic resonance (MR) imaging, the internal anal sphincter is typically slightly hyperintense on T2-weighted MR images. The external anal sphincter is contiguous with the levator ani and puborectalis muscles superiorly, is innervated by branches of the pudendal nerve and the perineal branch of the S4 nerve root, and contributes to voluntary control of fecal continence. The external sphincter is hypointense on T2-weighted MR images and can be easily recognized by its contrast in signal intensity from the surrounding high-signal-intensity intersphincteric fat plane medially and the ischioanal and ischiorectal fat laterally.
- The anal canal is tilted forward in the sagittal plane; thus, appropriate angulation of the oblique axial images orthogonal to the plane of the anal canal and of oblique coronal images parallel to the plane of the anal canal is essential for the proper assessment of the sphincter complex.
- Staging for anal cancer is based on the size of the tumor rather than the depth of invasion. Anal tumors smaller than 2 cm in greatest dimension are classified as stage T1 lesions, tumors measuring 2–5 cm as stage T2 lesions, and tumors larger than 5 cm as stage T3 lesions. Lesions that invade adjacent organs such as the vagina, urethra, prostate, or bladder are classified as stage T4 lesions. Rectal invasion is not considered stage T4 disease.
- In patients with a positive response to chemoradiation therapy, the size of the primary tumor decreases along with the signal intensity on T2-weighted MR images. In patients with a complete response, low signal intensity may be seen at the tumor site on T2-weighted MR images, a finding that indicates posttreatment fibrosis.
- A useful distinguishing imaging feature of aggressive angio-myxomas is the presence of a laminated appearance attributed to collagen fibrils in the background of a loose myxoid stroma.

HPV is sexually transmitted; the risk of cancer is greater in patients with a history of receptive anal intercourse. Anal cancer rates are greater in human immunodeficiency virus (HIV)-positive patients and in solid organ transplant recipients in whom cell-mediated immunity is suppressed, resulting in a poor immune response and the prolonged existence of HPV (3). In men who practice anal intercourse, the incidence of anal cancer is 35 per 100 000, and those who are HIV positive have twice the risk of those who are seronegative (3,5). Other risk factors for anal SCC include the coexistence of sexually transmitted diseases; a history of cervical, vulvar, or vaginal carcinoma; and cigarette smoking (4). It is no longer believed that chronic irritation of the perineum and the anal canal is a predisposing factor for the development of anal cancer (4). Less-frequent tumors include those associated with Bowen disease and Paget

disease, basal cell carcinoma, melanoma, and adenocarcinoma.

In this article, we review the World Health Organization classification of anal carcinoma, the imaging evaluation, and the treatment options for neoplasms of the anal canal and perianal skin.

Anatomy

The anal region is composed of the anal canal, the anal verge, and the anal margin. The anal canal extends from the dentate line, which is at the inferior margin of the anal columns of Morgagni, to the anal verge and measures approximately 4 cm. However, functionally accepted boundaries extend from the anorectal junction above, which is defined by the puborectalis muscle, to the anal verge below (6). The anal verge is a band of squamous epithelial tissue that lacks hair follicles and extends from the intersphincteric groove onto the perianal skin. The anal margin, or perianal skin, is identified by keratinized epithelium containing hair follicles and encompasses a radius of 5 cm from the anal verge. The muscles surrounding the anal canal control continence and can be separated into the internal anal sphincter (7) and the external anal sphincter (8). The internal anal sphincter (7) is the direct extension of the inner circular smooth muscle layer of the rectum and is innervated by both sympathetic and parasympathetic nerves, allowing for a state of tonic contraction and baseline continence (6). At magnetic resonance (MR) imaging, the internal anal sphincter is typically slightly hyperintense on T2-weighted MR images (Figs 1, 2) (9). The external anal sphincter (8) is contiguous with the levator ani and puborectalis muscles superiorly, is innervated by branches of the pudendal nerve and the perineal branch of the S4 nerve root, and contributes to voluntary control of fecal continence (10). The external sphincter is hypointense on T2-weighted MR images and can be easily recognized by its contrast in signal intensity from the surrounding high-signal-intensity intersphincteric fat plane medially and the ischioanal and ischiorectal fat laterally (Figs 1, 2) (9,11,12).

The dentate line divides the upper two-thirds and the lower one-third of the anal canal (Fig 2a) (3). The proximal one-third of the anal canal is made up of colorectal mucosa, and the middle third is made up of transitional zone mucosa; nonkeratinized squamous epithelium is found in the distal third. In addition, the anal canal contains scattered melanocytes and endocrine cells that can help explain the spectrum of tumor types that may arise in the anal canal (13). The dentate line also separates the patterns of lymphatic drainage of anal tumors; tumors above the dentate line drain into the perirectal

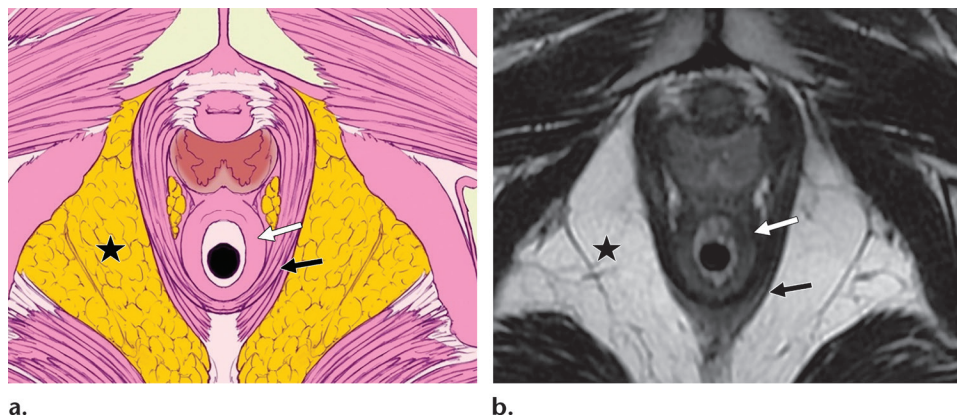


Figure 1. Anatomy of the anal canal in the axial plane. Drawing (a) and axial T2-weighted MR image (b) of the anatomic structures of the anal canal at the level of the ischiorectal fossa (★) show the external (black arrow) and internal (white arrow) anal sphincters. (Fig 1a reproduced, with permission, from Mayo Foundation for Medical Education and Research.)

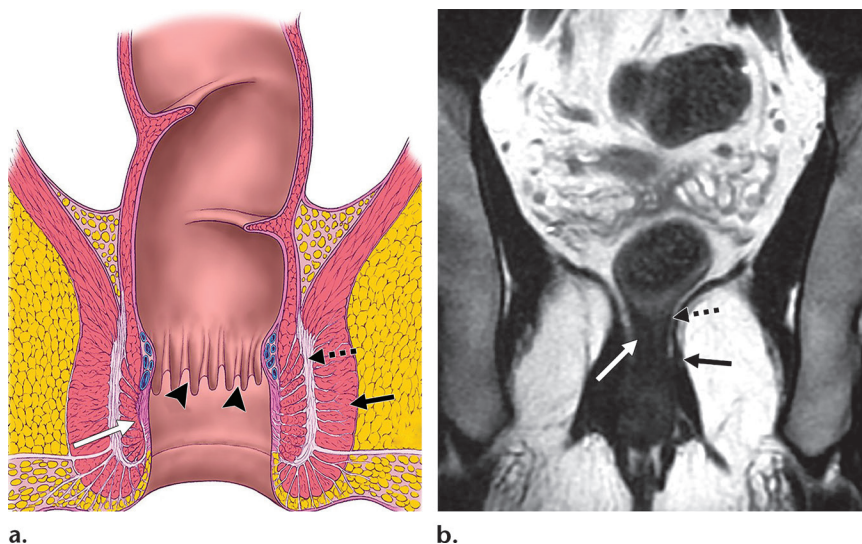


Figure 2. Anatomy of the anal canal in the coronal plane. (a) Drawing shows the external anal sphincter (solid black arrow), internal anal sphincter (white arrow), intersphincteric fat (dashed black arrow), and dentate line (arrowheads). (b) Coronal T2-weighted MR image of the anal canal shows the intersphincteric fat (dashed black arrow), the external anal sphincter (solid black arrow), which is contiguous with the puborectalis muscle, and the internal anal sphincter (white arrow), which is the contiguous extension of the inner circular muscle layer of the rectum. (Fig 2a reproduced, with permission, from Mayo Foundation for Medical Education and Research.)

and internal iliac nodes, whereas tumors below the dentate line and the anal margin drain into the inguinal and femoral lymph nodes (14).

MR Imaging Technique

MR imaging performed with surface coils provides excellent anatomic detail of the anal sphincters and the anatomic boundaries of the pelvis. Endo-anal coils provide superior anatomic resolution but have a limited field of view and are poorly tolerated in symptomatic patients. High-resolution T2-weighted MR images are the standard of reference in the evaluation of pathologic conditions of the anal canal (12). These images are ideally obtained

with 3-mm-thick sections by using a small field of view. The anal canal is tilted forward in the sagittal plane; thus, appropriate angulation of the oblique axial images orthogonal to the plane of the anal canal and of oblique coronal images parallel to the plane of the anal canal is essential for the proper assessment of the sphincter complex (15). The protocol used at our institution for the evaluation of anal disease includes oblique axial T1-weighted fast spin-echo MR pulse sequences, oblique axial and coronal T2-weighted fast spin-echo sequences, fat-saturated T2-weighted sequences that use short inversion time inversion recovery or spectrally adiabatic inversion recovery, and oblique axial and

Table 1: MR Imaging Protocol for Evaluating the Anal Canal and Perianal Region

MR Imaging Sequence	TR/TE (msec)	Section Thickness (mm)	Section Gap (mm)	No. of Signals Acquired
Coronal T2-weighted single-shot fast spin-echo imaging (to cover pelvis with a large FOV)	645/85	6	7	0.5
Axial, coronal, and sagittal (orthogonal to anal canal) T2-weighted fast-recovery fast spin-echo imaging	3000/100	3	0.3	4
Axial T2-weighted fast-recovery fast spin-echo fat-saturated imaging	4500/105	3	0.3	4
Axial nonenhanced and contrast-enhanced T1-weighted fat-saturated 3D volume gradient-echo imaging	450/2.5	3	0	2
Axial diffusion-weighted imaging ($b = 0, 400$, and 800 sec/mm^2)	3000/100	6	0	6

Note.—FOV = field of view, 3D = three-dimensional, TR/TE = repetition time/echo time.

oblique coronal fat-saturated T1-weighted fast spin-echo sequences with intravenous gadolinium-based contrast material. The three-dimensional isovoxel T2-weighted fast spin-echo sequence provides source data for postprocessing with image reformatting techniques to obtain reformatted images in any desired plane. Diffusion-weighted MR imaging is routinely performed for the evaluation of pathologic conditions of the anal canal (16). Commonly used b values include 0, 400, and 800 sec/mm^2 , followed by reconstruction of an apparent diffusion coefficient map (Table 1) (10,12).

Classification

Anal Canal Tumors

The current World Health Organization classification of tumors of the anal canal consists of intraepithelial and invasive tumors (Table 2) (4). Anal intraepithelial neoplasia is the precursor to anal SCC. Invasive tumors are further classified on the basis of cell type as primary epithelial tumors (SCC, adenocarcinoma, mucinous adenocarcinoma, small cell carcinoma, and undifferentiated carcinoma), nonepithelial tumors, carcinoid tumors, or melanoma. Secondary tumors (direct spread from rectal carcinoma, cervical carcinoma, or prostate carcinoma) may also involve the anal canal (3,4).

Anal Margin Tumors

Tumors in the anal margin are classified according to the World Health Organization classification of skin tumors (4). Anal margin tumors can also be intraepithelial or invasive. Anal intraepithelial neoplasia, also known as Bowen disease, is a precursor to SCC; and Paget disease is a precursor to adenocarcinoma (4). Invasive tumors include SCC, ad-

Table 2: World Health Organization Classification of Anal Canal Tumors

Tumor Type
Intraepithelial neoplasia
Anal intraepithelial neoplasia
Epithelial tumors
SCC
Adenocarcinoma
Mucinous adenocarcinoma
Small cell carcinoma
Undifferentiated carcinoma
Nonepithelial tumors
Leiomyoma
Gastrointestinal stromal tumor
Myofibroma
Carcinoid tumors
Melanoma
Secondary tumors
Direct extension from adjacent organs: rectal carcinoma, cervical carcinoma, and prostate carcinoma

Source.—Reference 4.

enocarcinoma, basal cell carcinoma, and verrucous carcinoma (Buschke-Löwenstein tumor), which is a rare variant of SCC (4).

Types of Anal Canal Tumors

Anal Intraepithelial Neoplasia

Anal intraepithelial neoplasia is a precursor lesion of anal SCC, and the incidence of anal intraepithelial neoplasia is increasing in immuno-

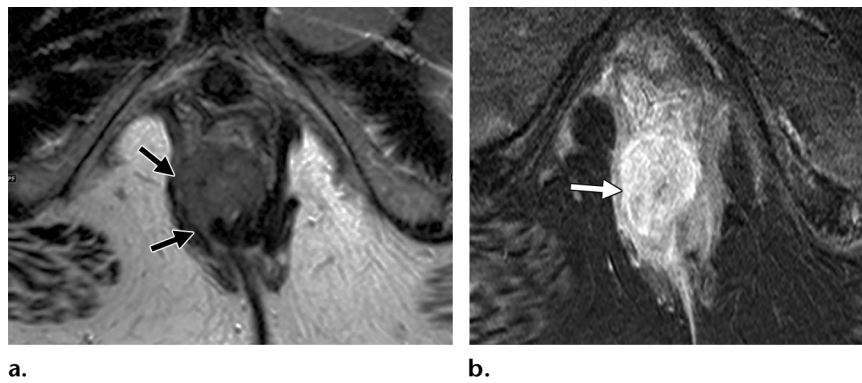


Figure 3. Anal SCC in a 63-year-old woman presenting with rectal bleeding. **(a)** Axial T2-weighted MR image shows a 4-cm infiltrative mass (arrows), a finding suggestive of a stage T2 tumor involving both the internal and external sphincters. **(b)** Axial gadolinium-enhanced T1-weighted MR image shows intense enhancement of the mass (arrow).

suppressed patients and those with HIV infection or HPV infection (17). Anal intraepithelial neoplasia is usually detected incidentally during the workup for other regional complaints, such as hemorrhoids (18). Anal intraepithelial neoplasia may appear as an eczematous, scaly, white or pigmented lesion with or without fissures. Induration or ulceration is a sign of poor prognosis and may indicate invasiveness. At histopathologic examination, lesions show a loss of stratification, dark nuclei, and nuclear pleomorphism. Anal intraepithelial neoplasia can be low grade or high grade, depending on the extent of the involved anal epithelium, and is treated with local ablation or surgery. HPV-based vaccines are also beneficial in the management of anal intraepithelial neoplasia in HPV-infected patients (19).

SCC of the Anal Canal

SCC is the most common type of anal canal cancer, accounting for 80%–85% of all anal canal cancers (20). SCC of the anal canal is increasingly diagnosed in patients with HIV infection and in immunosuppressed patients after transplantation. HPV infection, cigarette smoking, multiple sexual partners, and a history of previous or current intraepithelial neoplasia of the cervix and vulva are associated with a greater risk of developing anal canal SCC (20). Anal infection with HPV, in particular HPV types 16 and 18, is the most important risk factor in the development of anal SCC (3). SCC is associated with HPV infection in 90% of patients (21). In a large case-control study (22), investigators found that cigarette smoking is a substantial risk factor for the development of anal cancer. In addition, men with a history of receptive anal intercourse had a 33-fold greater relative risk for developing anal cancer (22). Homosexual men with HIV infection had double the risk for developing anal cancer compared with HIV-negative homosexual

men (8,22). The presenting symptoms include pain, ulceration, bleeding, discharge, and pruritus (20). MR imaging provides information about the tumor size, its local spread, and invasion of adjacent organs. At imaging, anal cancer appears as a lobulated intraluminal or infiltrative mass with a tendency to spread circumferentially. At MR imaging, anal carcinoma is isointense to hyperintense on T2-weighted MR images, isointense to hypointense on T1-weighted MR images, and enhances on contrast material-enhanced MR images (Figs 3, 4) (12).

Staging for anal cancer is based on the size of the tumor rather than the depth of invasion (1). Anal tumors smaller than 2 cm in greatest dimension are classified as stage T1 lesions, tumors measuring 2–5 cm as stage T2 lesions, and tumors larger than 5 cm as stage T3 lesions (Figs 3, 4). Lesions that invade adjacent organs such as the vagina, urethra, prostate, or bladder are classified as stage T4 lesions. Rectal invasion is not considered stage T4 disease (Table 3) (23). Regional lymph nodes include the perirectal, internal iliac, and inguinal nodes (1).

The current standard of care for treatment of anal canal SCC is definitive chemoradiation therapy (15). Chemoradiation therapy preserves the sphincter complex and yields an 80% survival rate at 5 years (15). Salvage surgery with abdominoperineal resection is reserved for only nonresponsive patients who have persistent or recurrent tumors after chemoradiation therapy (15). Assessment of the response to treatment after chemoradiation therapy for anal canal SCC is challenging, and recognition of the changes to be expected on MR images obtained after treatment is paramount to the interpretation. MR imaging after chemoradiation therapy often shows anal mucosal thickening and high T2 signal intensity that is due to edema. In patients with a positive response to

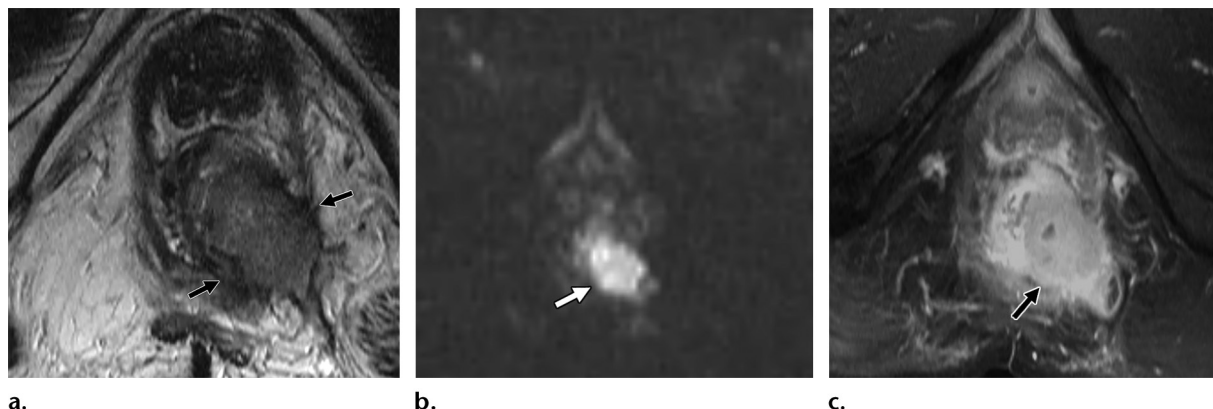


Figure 4. Anal SCC in a 58-year-old woman with a long history of HIV infection who presented with rectal bleeding. Axial T2-weighted (a), diffusion-weighted (b), and contrast-enhanced T1-weighted (c) MR images shows an ill-defined infiltrating anal mass (arrows) with involvement of the internal and external anal sphincters and extension to the skin. This mass shows restricted diffusion (arrow in b) and heterogeneous contrast enhancement (arrow in c).

Table 3: TNM Classification of Anal Canal Cancer

Descriptor	Definition
Primary tumor (T)	
Tx	Primary tumor cannot be assessed
Tis	Carcinoma in situ
T1	Tumor less than 2 cm in greatest dimension
T2	Tumor between 2 and 5 cm in greatest dimension
T3	Tumor more than 5 cm in greatest dimension
T4	Tumor invading adjacent organs
Regional lymph nodes (N)	
Nx	Regional nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in the perirectal nodes
N2	Metastasis in unilateral internal iliac and/or inguinal nodes
N3	Metastasis in bilateral internal iliac or inguinal nodes
Distant metastasis (M)	
M0	No distant metastasis
M1	Distant metastasis

Source.—Reference 23.

chemoradiation therapy, the size of the primary tumor decreases along with the signal intensity on T2-weighted MR images (15). In patients with a complete response, low signal intensity may be seen at the tumor site on T2-weighted MR images, a finding that indicates posttreatment fibrosis (12). Positron emission tomography (PET) with fluorine 18 fluorodeoxyglucose (FDG) integrated with computed tomography (CT) (FDG PET/CT) plays an important role in the evaluation of the treatment response in patients after chemoradiation therapy (24). The absence of residual metabolic activity at post-treatment FDG PET/CT has been shown to correlate with a longer progression-free survival time (25).

Adenocarcinoma of the Anal Canal

Anal adenocarcinoma accounts for 10%–15% of all anal canal cancers (26). Adenocarcinomas are thought to originate from the anal glands and then spread outward to involve the anal sphincter. Most anal adenocarcinomas are the result of distal spread of rectal adenocarcinoma (27). The clinical manifestation of adenocarcinoma is similar to that of SCC of the anal canal. Treatment strategies include total abdominoperineal resection, radiation therapy followed by tumor excision, and combined chemoradiation therapy (26,28–30). On T2-weighted MR images, mucinous adenocarcinomas may show a characteristic high signal intensity that might be less prominent in nonmucinous adenocarcinomas (31).

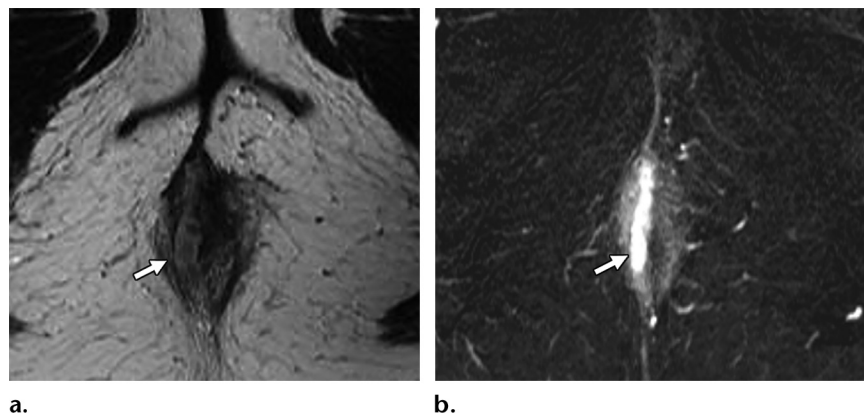


Figure 5. SCC within an anorectal fistula in a 68-year-old woman with a history of Crohn disease and a long-standing perianal fistula. (a) Coronal T2-weighted MR image shows a linear tract in the perianal region with heterogeneous internal content (arrow). (b) Coronal contrast-enhanced T1-weighted MR image shows enhancement of not only the tract lining, but also the internal content (arrow).

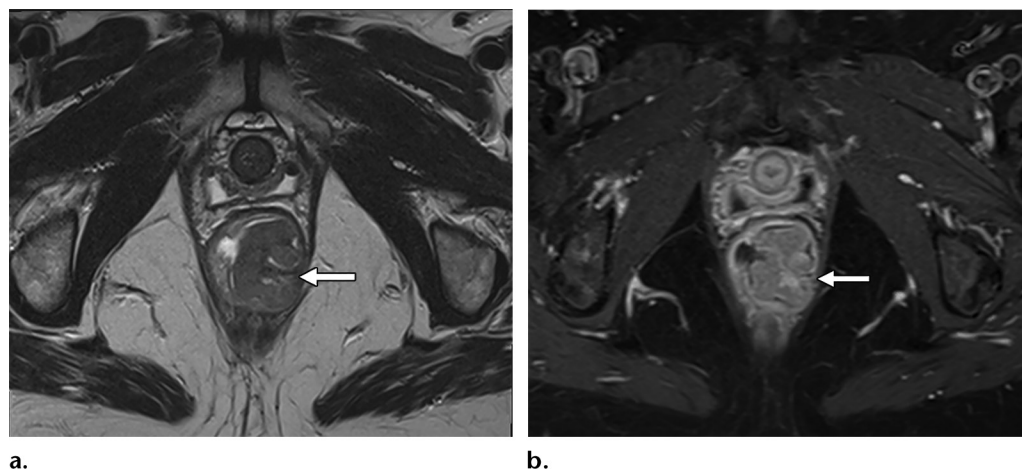


Figure 6. Anal melanoma in a 58-year-old woman. (a) Axial T2-weighted MR image shows a heterogeneously hyperintense anal mass (arrow). (b) Axial contrast-enhanced T1-weighted MR image shows heterogeneous enhancement of the mass (arrow).

Carcinoma within an Anorectal Fistula

Carcinoma arising from an anal fistula in patients with Crohn disease is rare (32). Tumors arising from nonhealing fistulas are either adenocarcinomas or SCCs (4). Carcinoma as a complication in patients with Crohn disease who have anogenital disease typically arises from a long-standing perianal fistula and manifests as advanced cancer at diagnosis (32). Symptoms are usually nonspecific, and the clinical or endoscopic examinations are often limited because of the patient's anal pain and stricture. Cross-sectional imaging has a low sensitivity in the detection of cancer complicating anorectal fistulas (32); thus, it is important for radiologists to have a high level of suspicion for possible neoplastic changes when they review the images of patients with perianal fistulas (32). The presence of enhancing soft tissue within the fistula should raise the possibility of malignant transformation (Fig 5) (33).

Anal Melanoma

Anal melanoma is rare, with an incidence of 0.5%–2% of all anorectal malignant neoplasms,

and accounts for only 2% of all melanomas (34). Anal melanoma usually manifests in elderly patients and has a slight female predominance (35). There is often a delay in the clinical diagnosis of anal melanomas because they are often misdiagnosed clinically as thrombosed hemorrhoids. The imaging features of anal melanomas depend on the melanin content and the presence or absence of hemorrhage. Melanocytic melanomas demonstrate both T1 and T2 shortening, whereas amelanotic melanomas typically do not exhibit substantial T1 or T2 shortening (Fig 6). The enhancement pattern is usually homogeneous (36). The primary approach to therapy is surgical, either with abdominoperineal resection or wide local excision (34). However, both of these procedures result in the same long-term outcomes (34). Adjuvant chemotherapy, immunotherapy, radiation therapy, or a combination of these plays a role in the management of advanced-stage anal melanomas (34).

Anal Lymphoma

The gastrointestinal tract is the most common extranodal location of non-Hodgkin lymphomas,

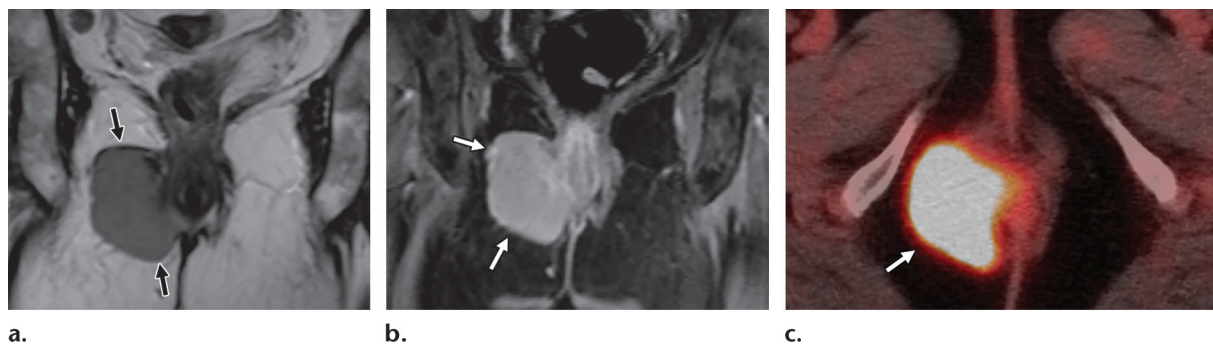


Figure 7. Anal lymphoma in a 64-year-old woman. (a, b) Coronal T2-weighted (a) and gadolinium-enhanced T1-weighted (b) MR images show a lobulated T2-hyperintense mass (arrows in a) that demonstrates intense contrast enhancement (arrows in b). (c) Axial FDG PET/CT image shows a mass with marked FDG activity (arrow). At histopathologic examination, the findings disclosed that this mass was a marginal zone B-cell mucosa-associated lymphoid tissue lymphoma.

with the stomach being the most commonly involved site, followed by the small intestine and the large intestine (37). Colorectal lymphoma accounts for 9%–20% of all gastrointestinal lymphomas but only 1% of colorectal malignant neoplasms (38,39). The most common histologic subtype is diffuse large B-cell non-Hodgkin lymphoma (40). Primary anorectal lymphoma is a rare entity, accounting for only 3% of all gastrointestinal lymphomas (39); the incidence is markedly higher in patients with AIDS. At imaging, anal lymphoma appears as a solid mass with focal and/or circumferential thickening of the anal canal, typically without luminal obstruction. At MR imaging, anal lymphoma lesions are isointense on nonenhanced T1-weighted MR images and hyperintense on T2-weighted MR images and show mild enhancement after administration of contrast material (Fig 7). FDG PET can be a useful diagnostic study because lymphomas show avid uptake of FDG (41).

Lymphangioma

Lymphatic malformations are congenital malformations of the lymphatic system, and approximately 75% of the lesions are located in the head and neck region. Intra-abdominal lymphatic malformations are usually found incidentally and account for nearly 5% of all lymphatic malformations. Intra-abdominal lymphatic malformations have been reported in the mesentery, gastrointestinal tract, spleen, liver, genitourinary system, and pancreas (42). Lymphatic malformations may become symptomatic if they become large enough to press on surrounding structures. Rare complications include intracystic hemorrhage, cyst rupture, and infection (43,44). MR imaging is the modality of choice for diagnosis of lymphangiomas. On MR images, a lymphatic malformation appears as a uni- or multicystic septate lesion with internal signal intensity characteristics that vary

depending on the contents of the cyst. Lymphangiomas that contain serous fluid demonstrate signal intensity characteristics similar to those of simple fluid, with low signal intensity on T1-weighted MR images and high signal intensity on T2-weighted MR images (Fig 8); whereas chylous and mixed-content lymphangiomas may show signal dropout on T1-weighted gradient-echo opposed-phase MR images compared with T1-weighted gradient-echo in-phase MR images and/or internal signal intensity suppression on fat-saturated MR images (45). The cyst wall and internal septa may show enhancement after administration of contrast material (Fig 8).

Myofibroma

Myofibroma is a benign neoplasm composed of myofibroblasts and can manifest as a solitary form or as a generalized form known as myofibromatosis (46). These tumors commonly occur in the skin and subcutaneous tissues of the head and neck. Solitary myofibromas of the gastrointestinal tract are rare, whereas as many as 40% of patients with myofibromatosis have associated visceral lesions (47,48). Solitary myofibromas of visceral organs have been described at various anatomic sites, including the pancreas, liver, testis, ovary, gastrointestinal tract, and brain (46). At MR imaging, myofibromas are isointense on T1-weighted MR images, are hypointense on T2-weighted MR images, and show intense enhancement after administration of contrast material (Fig 9). Myofibromas may also give the appearance of a “target sign” on contrast-enhanced images, with peripheral enhancement and a central hypointense area that reflects central tumor necrosis (46,48,49).

Leiomyoma

Leiomyomas of the anal canal are rare and may originate from the muscularis mucosae or muscularis propria. Anorectal leiomyomas are more

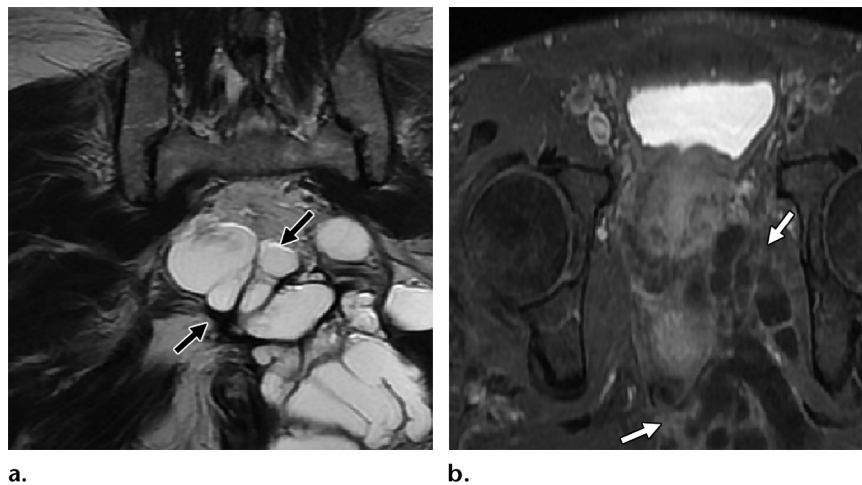


Figure 8. Perineal lymphatic malformation in a 70-year-old man with a long-standing history of a perineal and inguinoscrotal mass. **(a)** Coronal T2-weighted MR image shows a multiloculate cystic mass (arrows) extending and involving the perineal region, the pelvis, and the inguinoscrotal region. **(b)** Axial contrast-enhanced T1-weighted MR image shows peripheral enhancement of the cyst walls (arrows) without any internal enhancement.

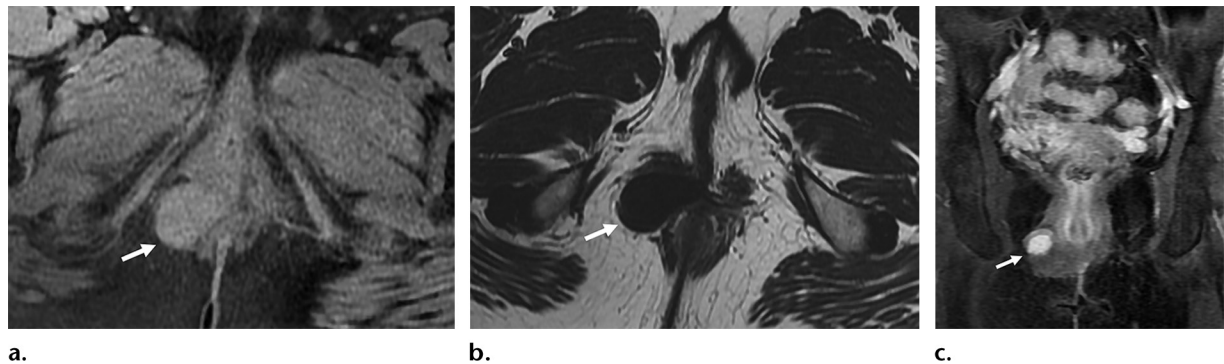


Figure 9. Perineal myofibroma in a 32-year-old woman. **(a)** Axial T1-weighted MR image shows a well-circumscribed right perineal mass (arrow) that is isointense to skeletal muscle. **(b)** Axial T2-weighted MR image shows a hypointense mass (arrow) and its origin from the external sphincter of the anal canal. **(c)** Coronal contrast-enhanced T1-weighted MR image shows intense enhancement of the mass (arrow) and its relationship with the external sphincter of the anal canal.

common in male subjects, with a peak incidence at the age of 60 years (50,51). At clinical presentation, patients with leiomyomas vary in their symptoms, and patients may be asymptomatic or may present with pelvic pain, a perineal mass, bleeding, dysuria, or difficult defecation (50). Smaller leiomyomas are confined to the anal canal, but larger leiomyomas may protrude through the anal canal and extend into the perineal region. Leiomyosarcoma is the malignant form of smooth muscle tumors. Histologically, leiomyosarcoma is differentiated from leiomyoma by marked necrosis, a high number of mitoses, large irregular nuclei, and deficient collagen stroma (50,52). At MR imaging, leiomyoma appears as a well-defined hypointense mass on T2-weighted MR images and is isointense on T1-weighted MR images (Fig 10) (53). Leiomyosarcomas are heterogeneous and isointense on T1-weighted MR images, are hyperintense on T2-weighted MR

images, and demonstrate heterogeneous enhancement on contrast-enhanced MR images (54). Surgical excision is the treatment of choice.

Gastrointestinal Stromal Tumor

Gastrointestinal stromal tumors of the anal canal are rare, accounting for 0.1%–0.4% of all gastrointestinal stromal tumors (55). Gastrointestinal stromal tumors arise from the interstitial cells of Cajal of the myenteric plexus, which control gastrointestinal tract motility. The biologic behavior of anal gastrointestinal stromal tumors is not clear, but most patients present with local disease, and less than 20% of patients present with or develop distant metastases (55). At MR imaging, a gastrointestinal stromal tumor appears as a large smoothly margined extraluminal mass arising from the anal canal. This tumor is hypo- to isointense on T1-weighted MR images and hyperintense on T2-weighted MR images.

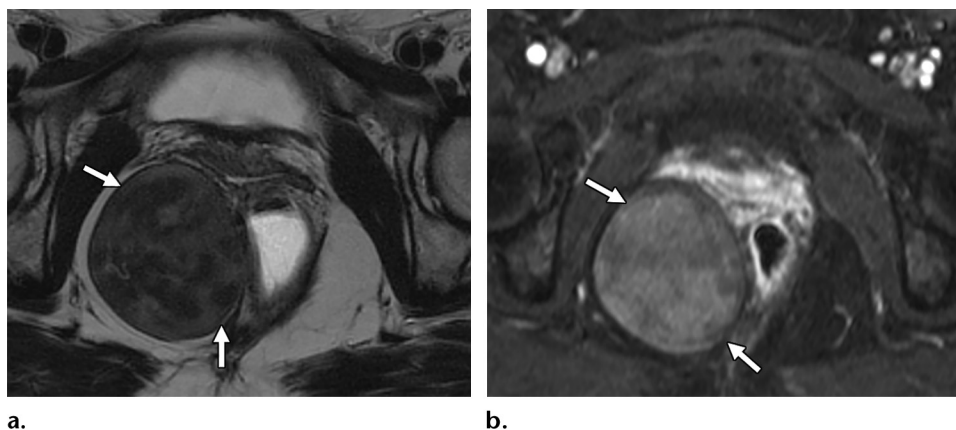


Figure 10. Leiomyoma of the perianal region in a 35-year-old woman. **(a)** Axial T2-weighted MR image shows a well-circumscribed mass (arrows) with characteristic T2 hypointensity in the right perianal region. **(b)** Axial contrast-enhanced T1-weighted MR image shows heterogeneous enhancement of the mass (arrows) and better depiction of the relationship with the anal canal.

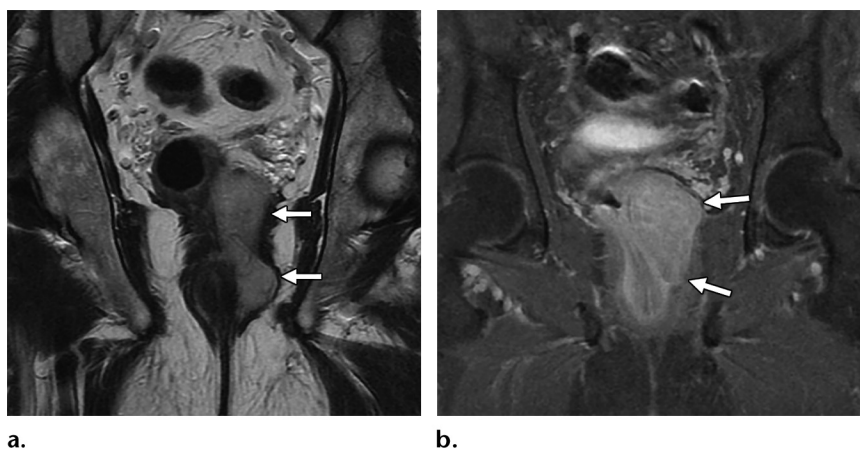


Figure 11. Gastrointestinal stromal tumor of the perianal region in a 52-year-old man. Coronal T2-weighted **(a)** and contrast-enhanced T1-weighted **(b)** MR images show a T2-hyperintense enhancing mural mass (arrows) in the left perianal region. At surgical resection, the mass proved to be a gastrointestinal stromal tumor.

At contrast-enhanced MR imaging, the pattern of enhancement is usually related to the tumor size; the larger the tumor, the more heterogeneous the enhancement is, which correlates with tumor necrosis (Fig 11) (56,57). The treatment of choice is wide local excision or abdominoperineal resection. The incidence of local recurrence is lower after abdominoperineal resection (55). However, the rate of distant metastasis and the survival rate after abdominoperineal resection are not significantly different from those after wide local excision (55). Adjuvant therapy or salvage therapy with imatinib mesylate, a tyrosine kinase inhibitor, has shown promising results in the treatment of patients with gastrointestinal stromal tumors (55).

Aggressive Angiomyxoma

Aggressive angiomyxomas are rare mesenchymal tumors that were first described in 1983 by

Steeper and Rosai (58). Aggressive angiomyxomas principally manifest in women of reproductive age, with a peak incidence in the 4th to 5th decades of life (59). Aggressive angiomyxomas usually involve the deep soft tissues of the vulvovaginal region, perianal region, and perineum of women and the inguinoscrotal region of men (59). Aggressive angiomyxomas tend to spread in multiple compartments without invading adjacent organs. Most aggressive angiomyxomas manifest as either a painless mass or a mass causing local pressure. A useful distinguishing imaging feature of aggressive angiomyxomas is the presence of a laminated appearance attributed to collagen fibrils in the background of a loose myxoid stroma (59) (Fig 12). The enhancement patterns of aggressive angiomyxomas vary. The laminated morphologic appearance can also be seen on contrast-enhanced CT and MR images (59).

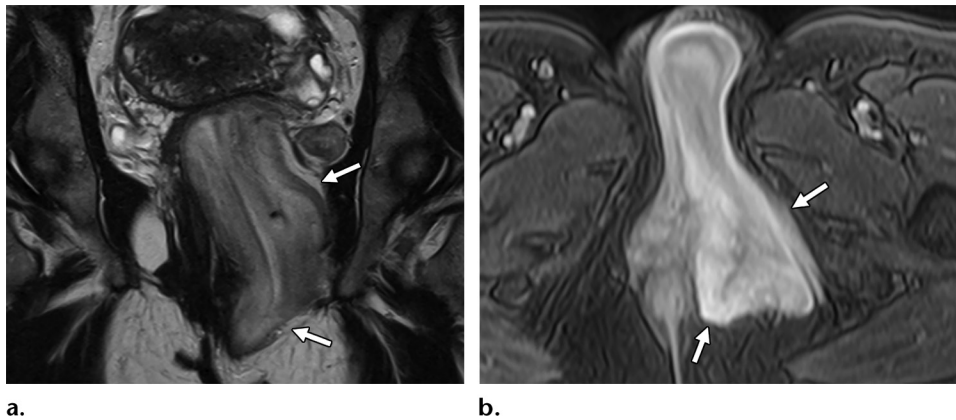


Figure 12. Aggressive angiomyxoma of the perianal region in a 35-year-old woman. **(a)** Coronal T2-weighted MR image shows a well-circumscribed hyperintense mass in the left perianal region, which crosses the urogenital diaphragm and extends into the left perirectal region and shows the characteristic laminated appearance (arrows). **(b)** Axial gadolinium-enhanced T1-weighted MR image shows diffuse enhancement of this mass (arrows). At surgical resection, this mass proved to be an aggressive angiomyxoma.

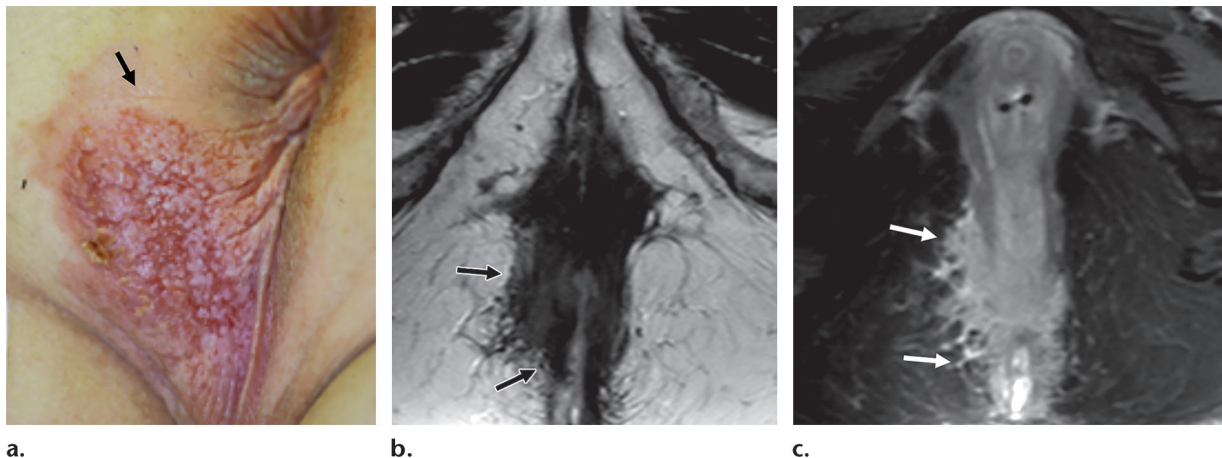


Figure 13. Paget disease of the anal margin in a 72-year-old woman. **(a)** Photograph of the anal margin shows an erythematous, eczematous, nodular plaquelike lesion (arrow) in the right perianal region. **(b, c)** Axial T2-weighted **(b)** and gadolinium-enhanced T1-weighted **(c)** MR images show an infiltrative hypointense mass lesion (arrows) involving the right anal margin; the mass demonstrates heterogeneous enhancement. (Fig 13a courtesy of Bashar Safar, MD, Department of Surgery, Johns Hopkins University, Baltimore, Md.)

Types of Anal Margin (Perianal) Tumors

Paget Disease (Intraepithelial Adenocarcinoma)

Perianal Paget disease is an intraepithelial adenocarcinoma and can be associated with underlying colorectal adenocarcinoma and anal carcinoma in 38%–69% of patients (60). Perianal Paget disease is associated with nonspecific symptoms such as perianal pain, bleeding, and itching; and the nonspecific symptoms frequently delay the diagnosis. Wide local excision is the treatment of choice, and abdominoperineal resection is considered for invasive disease. Invasive disease can metastasize to distant sites (61). MR imaging is useful for evaluating the extent and the depth of invasion (Fig 13). FDG

PET can be a useful diagnostic study because Paget disease may demonstrate FDG avidity at PET (62).

Bowen Disease (Intraepithelial SCC)

Perianal Bowen disease is considered a rare slow-growing premalignant lesion and is estimated to progress to invasive SCC in 2%–6% of cases (63). Bowen disease is associated with infection with HPV types 16 and 18. Common presenting symptoms include itching, burning, and bleeding. Wide local excision is the treatment of choice (63).

Verrucous Carcinoma

Verrucous carcinoma, also known as giant condyloma acuminatum or Buschke-Löwenstein tumor, is a large locally aggressive tumor with a

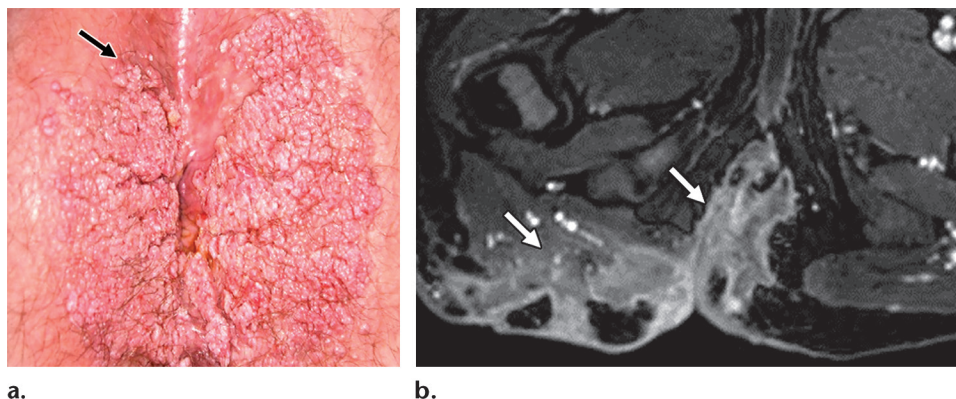


Figure 14. Verrucous carcinoma of the anal margin in a 44-year-old man with HIV infection and HPV infection. **(a)** Photograph of the perineum shows a large cauliflower-like anal mass (arrow). **(b)** Axial contrast-enhanced T1-weighted MR image shows extensive nodular thickening and enhancement of the perianal skin (arrows) that also extends to the right buttock region. (Fig 14a courtesy of Bashar Safar, MD, Department of Surgery, Johns Hopkins University, Baltimore, Md.)

cauliflower-like appearance (64). Giant condylomas are generally slow growing and are locally aggressive, with invasion of the surrounding tissue; however, distant metastases are rare (63). Verrucous carcinoma is strongly associated with HPV infection and is commonly seen in patients with HIV infection (63). Giant condyloma acuminatum has a malignant transformation rate as high as 50% (64). Treatment strategies include wide local excision; however, recurrence rates as high as 66% after wide local excision have prompted the use of neoadjuvant or adjuvant chemoradiation therapy to decrease the rate of recurrence (63,64). MR imaging is useful for evaluating the extent and the depth of invasion of these locally aggressive lesions (65–67). Condyloma acuminatum typically appears as an infiltrative heterogeneously enhancing mass that dissects into the perianal region. Lesions demonstrate intrinsic T2 hyperintensity relative to the surrounding musculature (Fig 14) (65–67).

Secondary Tumors

Anal involvement from malignancies arising from adjacent organs such as the rectum, cervix, prostate, and musculoskeletal system can occur and may mimic primary anal carcinoma (68,69). Rectal carcinomas involving the lower one-third of the rectum have a greater local recurrence rate and a higher risk of positive circumferential resection margins than do high-rectal and midrectal cancer, secondary to the tapering of the mesorectum and the proximity to the anal sphincter and the pelvic organs (70). It is imperative that radiologists thoroughly examine the tumor borders, the depth of extension, and the relation to the anal sphincters, to guide optimal surgical management. Surgical resection for rectal tumors includes total mesorectal

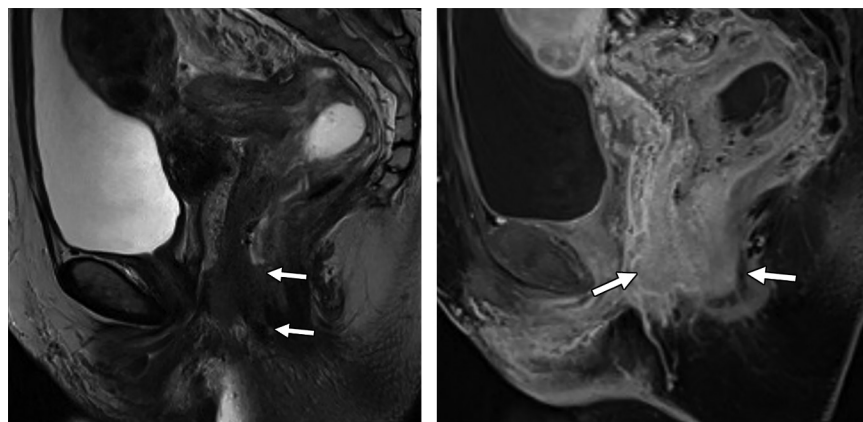
resection, either from a low anterior resection or from abdominoperineal resection, depending on the involvement of the anal sphincters (Fig 15) (71). Tumors away from the anal sphincters can be treated with low anterior resection, with resection of the tumor and with a continence-preserving anastomosis. Abdominoperineal resection is performed in patients with rectal tumors that involve the sphincter complex.

Prostate cancer can metastasize to the anal region by means of direct invasion or lymphatic spread (Fig 16) (72). Spread of cervical cancer to adjacent organs such as the urinary bladder, anal canal, and rectum constitutes stage IV disease. Cervical cancer usually appears as a hyperintense mass on T2-weighted MR images (73). Extension of cervical cancer to the anorectal mucosa might appear as segmental mucosal thickening or segmental disruption of the hypointense wall (Fig 17) (73).

Conclusion

A wide spectrum of tumors of the anal canal and perianal region either originate from the anal canal and anal margin or result from direct extension of tumors from adjacent organs. Although some tumors of the anal and perianal region have characteristic imaging findings that permit their diagnosis, tissue sampling is often warranted to confirm histopathologic findings, to establish a definitive diagnosis. Nevertheless, familiarity with the epidemiology, pathogenesis, imaging features, and treatment of these cancers can aid radiologic diagnoses and guide appropriate patient treatment.

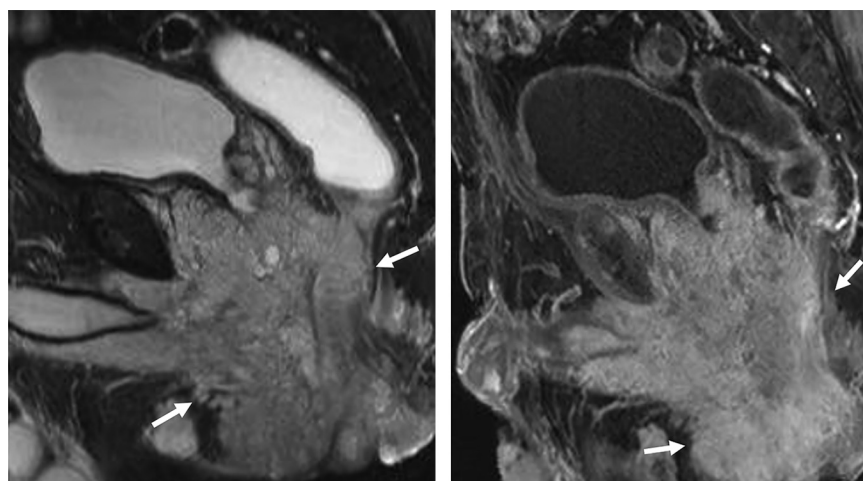
Acknowledgment.—We acknowledge the Media Department of the Mayo Clinic, Scottsdale, Ariz, for providing illustrations (Figs 1a and 2a).



a.

b.

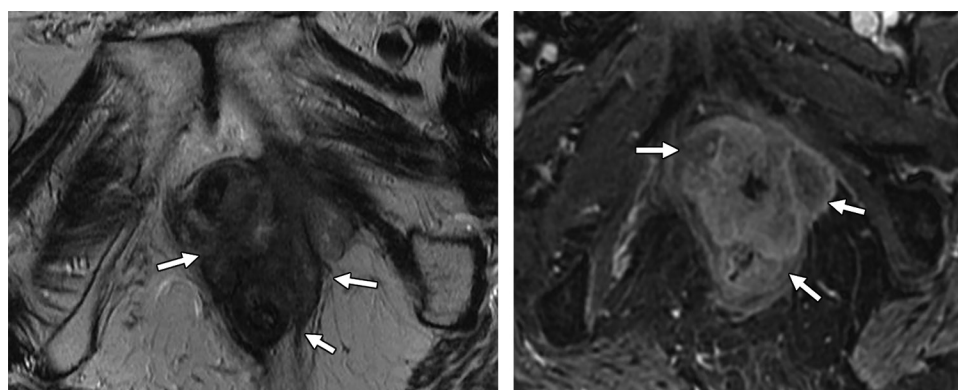
Figure 15. Rectal adenocarcinoma involving the lowest one-third of the rectum in a 45-year-old woman, with direct extension and involvement of the anal canal. Sagittal T2-weighted (a) and gadolinium-enhanced T1-weighted (b) MR images show an infiltrative mass (arrows) involving the lower rectum and anal canal, with heterogeneous contrast enhancement.



a.

b.

Figure 16. Adenoid cystic carcinoma of the prostate in a 71-year-old man with contiguous involvement of the anal canal. Sagittal T2-weighted (a) and contrast-enhanced T1-weighted (b) MR images show an infiltrative mass (arrows) centered in the region of the prostate, with extension to and involvement of the anal canal. The mass demonstrates heterogeneous contrast enhancement in b.



a.

b.

Figure 17. Stage IV cervical carcinoma in a 68-year-old woman with contiguous involvement of the anal canal. Axial T2-weighted (a) and contrast-enhanced T1-weighted (b) MR images show a large, heterogeneously enhancing infiltrating mass (arrows) invading the left pelvic sidewall, the urethra, and the anorectum, with involvement of the anal sphincters.

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